STAT452 60-Minute Written Test

Complete Mathematical Solutions

with R Verification

Applied Statistics for Engineers and Scientists II

Structure. Each problem is solved first with the mathematical notation used in the course Notes, followed by R code that verifies the calculation. The ridge result in Problem 4 is proved using derivatives and centered sums.

Based on `MidTerm_STAT452_23TT2.pdf` and Notes Chapters 1–4

Contents

Course-note alignment

This solution follows the course Notes directly:

- Chapter 1: the model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, least squares, R^2, and the partial F-test;
- Chapter 2: linearising an exponential model with $\log y$ and back-transforming to the original scale;
- Chapter 3: ridge penalties and the distinction between penalising and not penalising the intercept;
- Chapter 4: logistic probabilities, log-odds, odds ratios, and the threshold rule $\widehat{\pi} \geq 0.5$.

1 Problem 1: Multiple linear regression

For five students, let x_1 be self-study time, x_2 be tutoring time, and y be the test score. The assumed population model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

1.1 Part (a): Least-squares estimates and R-squared

Core idea

Least squares finds the plane whose vertical squared residuals have the smallest possible total. In matrix notation,

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$

The design matrix and response vector are

$$\boldsymbol{X} = \begin{bmatrix} 1 & 5 & 1 \\ 1 & 9 & 0 \\ 1 & 7 & 7 \\ 1 & 9 & 7 \\ 1 & 3 & 8 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 5 \\ 6 \\ 9.5 \\ 10 \\ 7.5 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{X}^\top \boldsymbol{X} = \begin{bmatrix} 5 & 33 & 23 \\ 33 & 245 & 141 \\ 23 & 141 & 163 \end{bmatrix}, \qquad \boldsymbol{X}^\top \boldsymbol{y} = \begin{bmatrix} 38 \\ 258 \\ 201.5 \end{bmatrix}.$$

Solving the normal equations $\boldsymbol{X}^\top \boldsymbol{X} \widehat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y}$ gives

$$\boxed{\widehat{\boldsymbol{\beta}} = \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 1.819205 \\ 0.486520 \\ 0.558644 \end{bmatrix}}$$

and hence

$$\boxed{\widehat{y} = 1.819205 + 0.486520 x_1 + 0.558644 x_2.}$$

Holding tutoring time fixed, one more hour of self-study is associated with an estimated 0.4865-point increase in score. Holding self-study time fixed, one more tutoring hour is associated with an estimated 0.5586-point increase.

Using the course-note definitions,

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = 0.281267, \qquad \text{SST} = \sum_{i=1}^{n}(y_i - \overline{y})^2 = 18.7.$$

Thus

$$\boxed{R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{0.281267}{18.7} = 0.984959.}$$

About 98.50% of the observed score variation in this five-student sample is explained by the two-predictor fitted model.

R verification

```
study <- data.frame(
  x1 = c(5, 9, 7, 9, 3),
  x2 = c(1, 0, 7, 7, 8),
  y  = c(5, 6, 9.5, 10, 7.5)
)

fit_full <- lm(y ~ x1 + x2, data = study)
X <- model.matrix(fit_full)
Y <- matrix(study$y, ncol = 1)

crossprod(X)                    # X'X
crossprod(X, Y)                 # X'y
solve(crossprod(X), crossprod(X,Y))

SSE <- sum(resid(fit_full)^2)
SST <- sum((study$y - mean(study$y))^2)
1 - SSE / SST
```

1.2 Part (b): Compare the reduced and full models

The reduced model is $Y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$. The full model adds x_2. The hypotheses are

$$H_0 : \beta_2 = 0 \qquad \text{versus} \qquad H_1 : \beta_2 \neq 0.$$

From Chapter 1, the partial F statistic is

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/(df_{E,R} - df_{E,F})}{\text{SSE}_F\ / df_{E,F}}.$$

Substituting the supplied ANOVA quantities,

$$F = \frac{(16.7941 - 0.2813)/(3-2)}{0.2813/2} = 117.42, \qquad p = 0.008409.$$

At $\alpha = 0.05$, $p < \alpha$, so reject H_0. Tutoring time supplies significant additional information after self-study time is already in the model. The more appropriate of the two candidates is

$$\boxed{Y \sim x_1 + x_2.}$$

```r
fit_reduced <- lm(y ~ x1, data = study)
anova(fit_reduced, fit_full)
```

1.3 Part (c): Prediction

For $x_1 = 5$ and $x_2 = 6$,

$$\begin{aligned}\widehat{y} &= 1.819205 + 0.486520(5) + 0.558644(6) \\ &= \boxed{7.603669}.\end{aligned}$$

```r
predict(fit_full, newdata = data.frame(x1 = 5, x2 = 6))
# 7.603669
```

Casio fx-580VN X check

In `MENU` $\rightarrow$ `Matrix`, enter $\boldsymbol{X}$ as MatA and $\boldsymbol{y}$ as MatB. Compute Trn(MatA)MatA and Trn(MatA)MatB, then evaluate

$$(\boldsymbol{X}^\mathsf{T}\boldsymbol{X})^{-1}\boldsymbol{X}^\mathsf{T}\boldsymbol{y}.$$

The displayed coefficient vector should be approximately

$$(1.819205, 0.486520, 0.558644)^\mathsf{T}.$$

2 Problem 2: Exponential regression

The proposed model is

$$Y_i = \exp(\beta_0 + \beta_1 x_i + \varepsilon_i), \qquad \varepsilon_i \sim N(0, \sigma^2).$$

2.1 Part (a): Linearisation and estimation

Taking natural logarithms produces a model that is linear in its parameters:

$$Z_i = \log Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The transformed observations are

x_i	1	2	3	4
y_i	2.98	4.65	5.00	6.50
$z_i = \log y_i$	1.091923	1.536867	1.609438	1.871802

For simple linear regression,

$$\widehat{\beta}_1 = \frac{S_{xz}}{S_{xx}}, \qquad \widehat{\beta}_0 = \overline{z} - \widehat{\beta}_1 \overline{x},$$

where

$$S_{xx} = \sum_{i=1}^{n}(x_i - \overline{x})^2, \qquad S_{xz} = \sum_{i=1}^{n}(x_i - \overline{x})(z_i - \overline{z}).$$

The estimates are

$$\boxed{\widehat{\beta}_0 = 0.924456, \qquad \widehat{\beta}_1 = 0.241221.}$$

Therefore,

$$\boxed{\log \widehat{y} = 0.924456 + 0.241221x.}$$

After back-transformation,

$$\widehat{y} = e^{0.924456} e^{0.241221x} = 2.52050(1.27280)^x.$$

Thus one additional year multiplies the fitted population by $e^{0.241221} = 1.27280$, an estimated increase of about 27.28%.

```r
population <- data.frame(
  year = 1:4,
  population = c(2.98, 4.65, 5.00, 6.50)
)
fit_exp <- lm(log(population) ~ year, data = population)
coef(fit_exp)
exp(coef(fit_exp)["year"])
```

2.2 Part (b): Predict the fifth year

$$\log \widehat{y}_5 = 0.924456 + 0.241221(5) = 2.130559,$$
$$\widehat{y}_5 = e^{2.130559} = \boxed{8.419576 \text{ million people.}}$$

This plug-in back-transformation estimates the conditional median on the original scale. Under an exact lognormal model, the conditional mean would be $\exp(\widehat{\eta} + \widehat{\sigma}^2/2)$; the standard exam answer is the requested plug-in prediction above.

```r
exp(predict(fit_exp, newdata = data.frame(year = 5)))
# 8.419576
```

Casio fx-580VN X check

Compute $\log y_i$ and use `MENU` $\to$ `Statistics` $\to$ `A+BX` with x_i and $\log y_i$. The calculator should return $a \approx 0.924456$ and $b \approx 0.241221$. Evaluate $\exp(a + 5b)$ to obtain 8.419576.

3 Problem 3: Logistic regression

Let

$$\pi(\boldsymbol{x}) = P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x}).$$

Following Chapter 4, logistic regression models the log-odds:

$$\operatorname{logit}\{\pi(\boldsymbol{x})\} = \log\left(\frac{\pi(\boldsymbol{x})}{1-\pi(\boldsymbol{x})}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

3.1 Part (a): Estimated model and interpretation

The fitted model is

$$\boxed{\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -3.449548 + 0.002294x_1 + 0.777014x_2 - 0.560031x_3.}$$

Equivalently, if

$$\widehat{\eta} = -3.449548 + 0.002294x_1 + 0.777014x_2 - 0.560031x_3,$$

then

$$\widehat{\pi} = \frac{e^{\widehat{\eta}}}{1+e^{\widehat{\eta}}} = \frac{1}{1+e^{-\widehat{\eta}}}.$$

Holding the other predictors fixed, a one-unit increase in x_j multiplies the odds of class 1 by $e^{\widehat{\beta}_j}$:

Predictor	$\widehat{\beta}_j$	$e^{\widehat{\beta}_j}$	Interpretation
x_1	0.002294	1.002297	odds increase by about 0.23%
x_2	0.777014	2.174968	odds are multiplied by about 2.175
x_3	-0.560031	0.571191	odds decrease by about 42.9%

All three reported slope p-values are below 0.05, so each predictor is statistically significant at the 5% level within this fitted model.

3.2 Part (b): Predicted probabilities and classes

Use the default threshold $\tau = 0.5$:

$$\widehat{Y} = \begin{cases} 1, & \widehat{\pi} \geq 0.5, \\ 0, & \widehat{\pi} < 0.5. \end{cases}$$

Row	$\widehat{\eta}$	$\widehat{\pi}$	$\widehat{\pi} \geq 0.5$	**Class**
1	-1.058484	0.257599	No	0
2	-0.940450	0.280809	No	0
3	0.362391	0.589619	Yes	1
4	-0.684196	0.335326	No	0

Hence the predicted classes are

$$\boxed{(0,0,1,0).}$$

```r
b <- c(-3.449548, 0.002294, 0.777014, -0.560031)
new <- data.frame(
  x1 = c(640, 600, 700, 620),
  x2 = c(3.35, 3.62, 3.56, 3.17),
  x3 = c(3, 3, 1, 2)
)

eta <- drop(cbind(1, as.matrix(new)) %*% b)
p <- plogis(eta)
cbind(new, eta, p, class = as.integer(p >= 0.5))
```

Casio fx-580VN X check

For each row, compute $\widehat{\eta}$ and then $1/(1+e^{-\widehat{\eta}})$. For row 1 the display should show $\widehat{\eta} \approx -1.05848$ and $\widehat{\pi} \approx 0.25760$, hence class 0.

4 Problem 4: Ridge estimator proof

What must be shown

For $\lambda > 0$, minimise

$$L(\beta_0, \beta_1) = \sum_{k=1}^{n} (y_k - \beta_0 - \beta_1 x_k)^2 + \lambda(\beta_0^2 + \beta_1^2).$$

Unlike the practical `glmnet` model, this question explicitly penalises both the intercept and slope.

Define the centered sums

$$S_{xx} = \sum_{k=1}^{n} (x_k - \overline{x})^2, \qquad S_{xy} = \sum_{k=1}^{n} (x_k - \overline{x})(y_k - \overline{y}).$$

Proof strategy

Differentiate L with respect to both parameters. Solve the first normal equation for $\widehat{\beta}_0$, substitute it into the second, and rewrite the uncentered sums using S_{xx} and S_{xy}.

Proof. Differentiating with respect to β_0 gives

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{k=1}^{n} (y_k - \beta_0 - \beta_1 x_k) + 2\lambda\beta_0.$$

At the minimiser, this derivative is zero, so

$$\begin{aligned} 0 &= -\sum_{k=1}^{n} y_k + n\widehat{\beta}_0 + \widehat{\beta}_1 \sum_{k=1}^{n} x_k + \lambda\widehat{\beta}_0 \\ &= -n\overline{y} + (n+\lambda)\widehat{\beta}_0 + n\overline{x}\widehat{\beta}_1. \end{aligned}$$

Therefore,

$$\boxed{\widehat{\beta}_0 = \frac{n}{n+\lambda}(\overline{y} - \widehat{\beta}_1 \overline{x}).} \tag{1}$$

Next,

$$\frac{\partial L}{\partial \beta_1} = -2\sum_{k=1}^{n} x_k(y_k - \beta_0 - \beta_1 x_k) + 2\lambda\beta_1.$$

Setting this equal to zero yields

$$\sum_{k=1}^{n} x_k y_k = n\overline{x}\widehat{\beta}_0 + \widehat{\beta}_1\left(\sum_{k=1}^{n} x_k^2 + \lambda\right). \tag{2}$$

Substitute (1) into (2):

$$\sum_{k=1}^{n} x_k y_k = n\overline{x}\frac{n}{n+\lambda}(\overline{y} - \widehat{\beta}_1\overline{x}) + \widehat{\beta}_1\left(\sum_{k=1}^{n} x_k^2 + \lambda\right).$$

Collecting the $\widehat{\beta}_1$ terms gives

$$\widehat{\beta}_1\left[\sum_{k=1}^{n} x_k^2 + \lambda - \frac{n^2}{n+\lambda}\overline{x}^{,2}\right] = \sum_{k=1}^{n} x_k y_k - \frac{n^2}{n+\lambda}\overline{x}\,\overline{y}. \tag{3}$$

Use the identities

$$\sum_{k=1}^{n} x_k y_k = S_{xy} + n\overline{x}\,\overline{y}, \qquad \sum_{k=1}^{n} x_k^2 = S_{xx} + n\overline{x}^{,2}.$$

The right-hand side of (3) becomes

$$S_{xy} + n\overline{x}\,\overline{y} - \frac{n^2}{n+\lambda}\overline{x}\,\overline{y} = S_{xy} + \frac{n\lambda}{n+\lambda}\overline{x}\,\overline{y}.$$

The bracket on the left-hand side becomes

$$\begin{aligned} S_{xx} + n\overline{x}^{,2} + \lambda - \frac{n^2}{n+\lambda}\overline{x}^{,2} &= S_{xx} + \lambda + \frac{n\lambda}{n+\lambda}\overline{x}^{,2} \\ &= S_{xx} + \lambda\left(1 + \frac{n}{n+\lambda}\overline{x}^{,2}\right). \end{aligned}$$

Consequently,

$$\boxed{\widehat{\beta}_1 = \frac{S_{xy} + \dfrac{n\lambda}{n+\lambda}\overline{x}\,\overline{y}}{S_{xx} + \lambda\left(1 + \dfrac{n}{n+\lambda}\overline{x}^{,2}\right)}.} \tag{4}$$

Equations (1) and (4) are the required results. □

Because L is a strictly convex quadratic for $\lambda > 0$, its Hessian is positive definite; therefore the stationary point above is the unique global minimiser.

R verification of the proof

The following code compares the derived formulas with the penalised matrix solution. Here the 2×2 identity matrix penalises both coefficients.

```
x <- c(1, 2, 4, 5)
y <- c(2, 3, 7, 8)
lambda <- 2
n <- length(y)
x_bar <- mean(x); y_bar <- mean(y)
Sxx <- sum((x - x_bar)^2)
Sxy <- sum((x - x_bar) * (y - y_bar))

b1 <- (Sxy + n*lambda*x_bar*y_bar/(n+lambda)) /
      (Sxx + lambda*(1 + n*x_bar^2/(n+lambda)))
b0 <- n*(y_bar - b1*x_bar)/(n+lambda)

X <- cbind(1, x)
matrix_answer <- solve(crossprod(X) + lambda*diag(2),
                       crossprod(X, y))
rbind(derived = c(b0, b1), matrix = matrix_answer)
#          [,1] [,2]
# derived  0.333  1.5
# matrix   0.333  1.5
```

Final answer summary

Item	Answer
1(a)	$\widehat{y} = 1.819205 + 0.486520x_1 + 0.558644x_2$, $R^2 = 0.984959$
1(b)	Use the full model; $F = 117.42$, $p = 0.008409$
1(c)	$\widehat{y}(5, 6) = 7.603669$
2(a)	$\log \widehat{y} = 0.924456 + 0.241221x$
2(b)	$\widehat{y}_5 = 8.419576$ million
3(a)	$\text{logit}(\widehat{\pi}) = -3.449548 + 0.002294x_1 + 0.777014x_2 - 0.560031x_3$
3(b)	Predicted classes $(0, 0, 1, 0)$
4	Equations (1) and (4)